

DA3-CAD

Multi-view RGB → auditable geometry → editable B-Rep

01

RGB + target



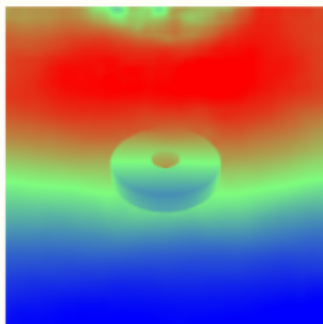
12 views · one selected object

10

objects

02

DA3



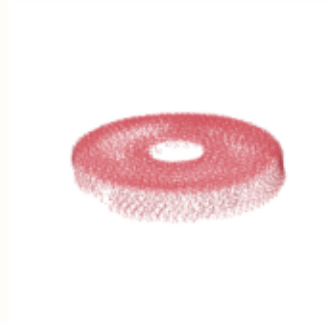
depth · confidence · K · E

120

RGB views

03

Audited 3D



pose gate · observed · trusted

10

valid STEP

04

CAD grammar

Sketch
line · circle

Operations
extrude · cut
revolve · shell

sketch · cut · extrude · revolve

10

product accepts

05

Editable B-Rep



CadQuery · STEP · STL · parameters

0

safe abstentions

No named-part classifier. No hidden primitive fallback. Unsupported evidence becomes ABSTAIN.

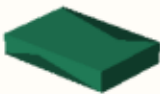
Research alpha · visible topology and concave profiles remain open problems

Public photo → CAD benchmark v2

10 STEP · 10 product accepts · 0 abstain

10 project-generated objects · 120 RGB views · DA3-LARGE-1.1 · evaluator-only reference CAD

Solid blockACCEPT



IoU 96.27% CD²×1000 0.179

extrude · through-holes 0/0

Hexagonal prismACCEPT



IoU 93.72% CD²×1000 0.284

extrude · through-holes 0/0

L bracketACCEPT



IoU 79.71% CD²×1000 0.492

extrude · through-holes 0/0

T profileACCEPT



IoU 71.79% CD²×1000 1.252

extrude · through-holes 0/0

U channelACCEPT



IoU 77.69% CD²×1000 0.730

extrude · through-holes 0/0

Key plateACCEPT



IoU 88.06% CD²×1000 0.274

extrude · through-holes 1/1

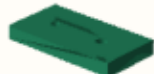
Round flangeACCEPT



IoU 89.75% CD²×1000 0.296

extrude · through-holes 1/1

Two-hole plateACCEPT



IoU 88.95% CD²×1000 0.300

extrude · through-holes 2/2

Stepped shaftACCEPT



IoU 88.70% CD²×1000 0.356

revolve · through-holes 0/0

Bottle profileACCEPT



IoU 91.47% CD²×1000 0.498

revolve · through-holes 0/0

Measured table · no hidden exclusions

IoU and Chamfer are evaluator-only; product did not see reference CAD

Object	Decision	Family	IoU %	CD²×1000	Holes	Failure / boundary
Solid block	accept	extrude	96.27	0.179	0/0	—
Hexagonal prism	accept	extrude	93.72	0.284	0/0	—
L bracket	accept	extrude	79.71	0.492	0/0	—
T profile	accept	extrude	71.79	1.252	0/0	—
U channel	accept	extrude	77.69	0.730	0/0	—
Key plate	accept	extrude	88.06	0.274	1/1	—
Round flange	accept	extrude	89.75	0.296	1/1	—
Two-hole plate	accept	extrude	88.95	0.300	2/2	—
Stepped shaft	accept	revolve	88.70	0.356	0/0	—
Bottle profile	accept	revolve	91.47	0.498	0/0	—